

FermiLab is working to understand our universe by studying high-energy particles. Their deep learning model, NuGraph3, aims to identify the resulting particles from neutrino particle interaction data. The clinic team worked to improve efficiency and accuracy of particle clustering, a main step in particle identification. Neutrinos are invisible to modern detectors; classifying their interactions is a big step towards physical understanding of the world.

Currently, model training can take up to four days. The team improved the efficiency of the model by enabling it to run on multiple GPUs at once, where previous versions could only run on one GPU. They also changed the particle clustering software to a CPU version, which frees GPUs for computationally expensive model training. These changes decreased training time to under one day.

To improve accuracy, FermiLab needs to understand when the model fails. The team built an interactive, GPU-supported dashboard showing true and predicted particle labels for each interaction individually. Using the dashboard, they understood that the current clustering algorithm is not sufficient to separate particles in complex interactions. They used this knowledge to update the algorithm, leading to improved clustering for some interactions. Future work will involve further updates to improve clustering overall.

Event Displays - NuGraph3

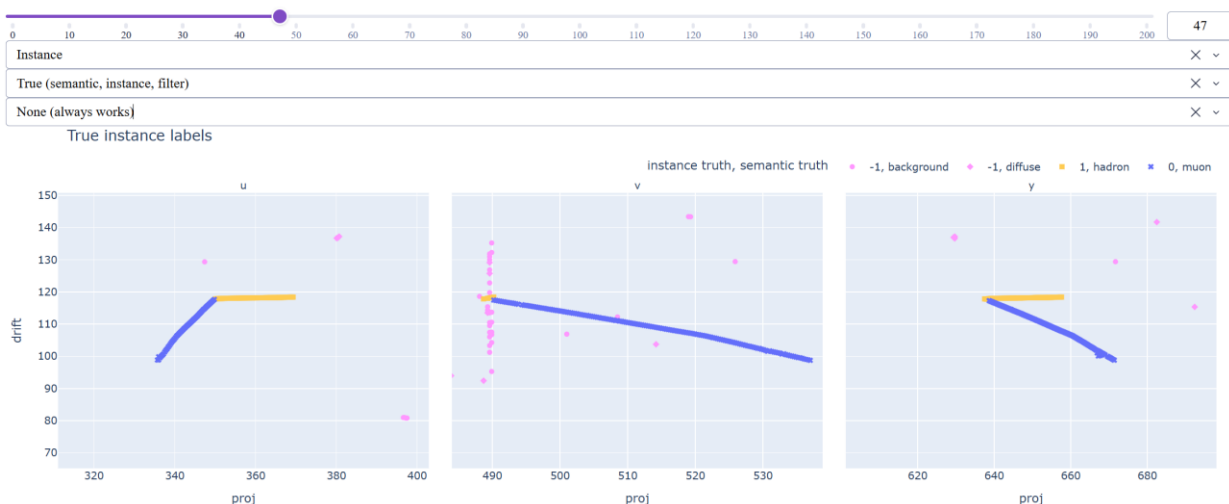


Figure 1: Web Browser View of NuGraph3 Interactive Dashboard. Browser shows true and predicted particle labels for one neutrino interaction (event) at a time from three different angles. Researchers can specify which interaction and what kind of particle labels they want to see, along with whether to see true or predicted labels and how to filter the view.